

Phagocytosis of Glioma Cells Enhances the Immunosuppressive Phenotype of Bone Marrow-Derived Macrophages.

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Tumor-associated macrophages (TAM) play a crucial role in immunosuppression. However, how TAMs are transformed into immunosuppressive phenotypes and influence the tumor microenvironment (TME) is not fully understood. Here, we utilized single-cell RNA sequencing and whole-exome sequencing data of glioblastoma (GBM) tissues and identified a subset of TAMs dually expressing macrophage and tumor signatures, which were termed double-positive TAMs. Double-positive TAMs tended to be bone marrow-derived macrophages (BMDM) and were characterized by immunosuppressive phenotypes. Phagocytosis of glioma cells by BMDMs in vitro generated double-positive TAMs with similar immunosuppressive phenotypes to double-positive TAMs in the GBM TME of patients. The double-positive TAMs were transformed into M2-like macrophages and drove immunosuppression by expressing immune-checkpoint proteins CD276, PD-L1, and PD-L2 and suppressing the proliferation of activated T cells. Together, glioma cell phagocytosis by BMDMs in the TME leads to the formation of double-positive TAMs with enhanced immunosuppressive phenotypes, shedding light on the processes driving TAM-mediated immunosuppression in GBM. Bone marrow-derived macrophages phagocytose glioblastoma cells to form double-positive cells, dually expressing macrophage and tumor signatures that are transformed into M2-like macrophages and drive immunosuppression.

Advancing precision medicine in gliomas through single-cell sequencing: unveiling the complex tumor microenvironment.

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Glioblastoma (GBM) displays an infiltrative growth characteristic that recruits neighboring normal cells to facilitate tumor growth, maintenance, and invasion into the brain. While the blood-brain barrier serves as a critical natural defense mechanism for the central nervous system, GBM disrupts this barrier, resulting in the infiltration of macrophages from the peripheral bone marrow and the activation of resident microglia. Recent advancements in single-cell transcriptomics and spatial transcriptomics have refined the categorization of cells within the tumor microenvironment for precise identification. The intricate interactions and influences on cell growth within the tumor microenvironment under multi-omics conditions are succinctly outlined. The factors and mechanisms involving microglia, macrophages, endothelial cells, and T cells that impact the growth of GBM are individually examined. The collaborative mechanisms of tumor cell-immune cell interactions within the tumor microenvironment synergistically promote the growth, infiltration, and metastasis of gliomas, while also influencing the immune status and therapeutic response of the tumor microenvironment. As immunotherapy continues to progress, targeting the cells within the inter-tumor microenvironment emerges as a promising novel therapeutic approach for GBM. By comprehensively understanding and intervening in the intricate cellular interactions within the tumor microenvironment, novel therapeutic modalities may be developed to enhance treatment outcomes for patients with GBM.

Microglia and macrophages in glioblastoma: landscapes and treatment directions.

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Glioblastoma is the most common primary malignant tumour of the central nervous system and remains uniformly and rapidly fatal. The tumour-associated macrophage (TAM) compartment comprises brain-resident microglia and bone marrow-derived macrophages (BMDMs) recruited from the periphery. Immune-suppressive and tumour-supportive TAM cell states predominate in glioblastoma, and immunotherapies, which have achieved striking success in other solid tumours have consistently failed to improve survival in this 'immune-cold' niche context. Hypoxic and necrotic regions in the tumour core are found to enrich, especially in anti-inflammatory and immune-suppressive TAM cell states. Microglia predominate at the invasive tumour margin and express pro-inflammatory and interferon TAM cell signatures. Depletion of TAMs, or repolarisation towards a pro-inflammatory state, are appealing therapeutic strategies and will depend on effective understanding and classification of TAM cell ontogeny and state based on new single-cell and spatial multi-omic in situ profiling. Here, we explore the application of these datasets to expand and refine TAM characterisation, to inform improved modelling approaches, and ultimately underpin the effective manipulation of function.

Macrophages in Glioblastoma Development and Therapy: A Double-Edged Sword.

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Glioblastoma (GBM) is one of the leading lethal tumors, featuring aggressive malignancy and poor outcome to current standard temozolomide (TMZ) or radio-based therapy. Developing immunotherapies, especially immune checkpoint inhibitors, have improved patient outcomes in other solid tumors but remain fatigued in GBM patients. Emerging evidence has shown that GBM-associated macrophages (GAMs), comprising brain-resident microglia and bone marrow-derived macrophages, act critically in boosting tumor progression, altering drug resistance, and establishing an immunosuppressive environment. Based on its crucial role, evaluations of the safety and efficacy of GAM-targeted therapy are ongoing, with promising (pre)clinical evidence updated. In this review, we summarized updated literature related to GAM nature, the interplay between GAMs and GBM cells, and GAM-targeted therapeutic strategies.

Expression of GATA6 in the human and mouse central nervous system.

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The mammalian GATA family of transcription factors comprises of 6 members that are involved in diverse roles. The expression profile of GATA6 has been poorly defined in the central nervous system (CNS). In this report, we identify GATA6 expression in the normal mouse and human CNS, using Northern blot analyses, immunohistochemistry (IHC), and immunofluorescence (IF). GATA6 is expressed as a 2.2 kb transcript in the adult mouse brain and several regions of the adult human brain. Furthermore, cellular characterization demonstrates GATA6 nuclear expression in neurons, astrocytes, choroids plexus epithelium, and endothelial cells.

GATA-4 regulates neuronal apoptosis after intracerebral hemorrhage via the

NF- κ B/Bax/Caspase-3 pathway both in vivo and in vitro.

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GATA-binding protein 4 (GATA-4) is a member of the GATA family of transcription factors, is expressed in the normal brain and participates in the neural inflammatory response and senescence. However, few studies have investigated whether GATA-4 is involved in the brain damage induced by intracerebral hemorrhage (ICH). The aim of this study was to investigate in vivo and in vitro the role of GATA-4 in ICH-induced secondary brain injury (SBI) and its potential underlying mechanisms. A rat model of ICH was established by autologous blood injection in vivo. In vitro, oxidized hemoglobin was applied to mimic the effects of ICH in neuronal culture. The function of GATA-4 and its mechanism of action after ICH were investigated using siRNA-mediated knockdown and plasmid-mediated overexpression techniques combined with immunofluorescence, western blot, and other molecular methods. It was found that the expression of GATA-4 was increased in the brain of rats after ICH, and its phosphorylation also increased correspondingly. Furthermore, knocking down the expression of GATA-4 led to a significant decrease in neurobehavioral scores and neuronal apoptosis, indicating that secondary brain damage was improved. Conversely, the overexpression of GATA-4 aggravated brain damage. Blockade of a critical phosphorylation site on the GATA-4 overexpression plasmid alleviated the exacerbated damage in vitro and in vivo. Moreover, GATA-4 promoted the activation of NF- κ B, and increased the expression of Bax, and cysteine aspartate-specific protease 3 (caspase-3) in its cleaved form, causing neuronal apoptosis. In conclusion, the expression of GATA-4 was increased in the brain of rats after ICH. GATA-4 phosphorylation mediates the function of the protein in ICH-induced SBI. Neuronal apoptosis after ICH was mainly induced by NF- κ B activation, which was promoted by GATA-4.

GATA4 is a regulator of astrocyte cell proliferation and apoptosis in the human and murine central nervous system.

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The GATA transcription factors consist of six family members, which bind to the consensus DNA-binding element, W-GATA-R, and are poorly characterized in the central nervous system (CNS). Using retroviral gene trapping on transgenic mouse glioma models, we identified GATA6 to be a novel tumor suppressor gene in glioblastoma multiforme. We now show GATA4, a family member of GATA6, to be expressed in the neurons and glia of normal murine and human embryonic and adult CNS. Silencing GATA4 in normal astrocytes did not alter their growth properties. In contrast, knockdown of Gata4 in p53 null non-transformed murine astrocytes induced transformation, with increased proliferation and resistance to chemotherapy or radiation-induced apoptosis. Furthermore, GATA4 expression was lost in a panel of human malignant astrocytoma cell lines. GATA4 overexpression in normal human and murine astrocytes resulted in a cell cycle block in G1 phase, with increased apoptosis. Mechanistically, GATA4 was a transcriptional inducer of the cyclin-dependent kinase inhibitor, p15^{INK4B}, leading to the attenuation of cyclin D1. GATA4 expression was also induced by transforming growth factor-beta, leading to the inhibition of astrocyte proliferation. Collectively, we show that GATA4 is expressed in the embryonic and adult CNS and acts as a negative regulator of astrocyte proliferation and growth.

Temporal and spatial changes in GATA transcription factor expression are coincident

with development of the chicken optic tectum.

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The molecular mechanisms specifying patterns of gene expression in the vertebrate brain, which in turn determine the developmental fates of specific neurons, are yet to be clearly defined. Individual members of a recently identified family of transcriptional regulatory proteins, the GATA factors, are required for the differentiation of certain hematopoietic cell lineages. We show here that two of the members of this gene family, GATA-2 and GATA-3, are expressed within discrete cell populations of the chicken optic tectum during embryogenesis, and that they have highly restricted patterns of expression in the developing chicken brain. Furthermore, the induction of GATA factor expression within specific cell layers parallels the well established spatial (rostral to caudal) and temporal pattern of optic tectum development. The observation that both the timing of appearance and the localization of expression of GATA-2 and GATA-3 are correlated with optic tectum development suggest that these transcription factors may be associated with the initiation of gene transcription required for the determination of specific neuronal fates within visual areas of the vertebrate brain.

GATA factors differentially activate multiple gonadal promoters through conserved GATA regulatory elements.

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The GATA factors are a group of transcriptional regulators that play essential roles in cell differentiation, organ morphogenesis, and tissue-specific gene expression during development. The six vertebrate GATA factors are expressed in a broad spectrum of tissues, including the hemopoietic system, heart, gut, brain, placenta, pituitary, and gonads. Interestingly, GATA-like DNA-binding proteins are found in the gonads of several species, ranging from lower invertebrates to humans, thus supporting an evolutionary conserved and crucial role for these factors in gonadal development and function. Indeed, GATA factors are expressed from the onset of gonadal development and are later found in multiple cell lineages of both the testis and ovary. We now report that GATA-4 differentially activates transcription of several genes expressed in the gonads that encode either steroidogenic enzymes (steroidogenic acute regulatory protein and aromatase), hormones (inhibin alpha and Müllerian inhibiting substance) and a transcription factor (SF-1) known to be essential for gonadal development and function. Thus, our results identify GATA-4 as an important regulator of gonadal gene transcription where its specificity of action is mediated through synergistic interactions with other transcription factors such as SF-1.

GATA transcription factors in development and disease.

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The GATA family of transcription factors is of crucial importance during embryonic development, playing complex and widespread roles in cell fate decisions and tissue morphogenesis. GATA proteins are essential for the development of tissues derived from all three germ layers, including the skin, brain, gonads, liver, hematopoietic, cardiovascular and urogenital systems. The crucial activity of GATA factors is underscored by the fact that inactivating mutations in most GATA members lead to embryonic lethality in mouse models and are often associated with developmental diseases in humans. In this Primer, we

discuss the unique and redundant functions of GATA proteins in tissue morphogenesis, with an emphasis on their regulation of lineage specification and early organogenesis.
